

Bending free energy with renormalized area
OR

How Safran does it

OR

What is δ ?

For small H , we can expand the elastic free energy like so:

$$f_{el} = f_0 + f_1 H + f_2 H^2 + o(H^3)$$

even if $c_0 \neq 0$. However, f_0, f_1, f_2 could be dependent on Σ . Let's assume they are and see what we get:

$\Delta\Sigma = \Sigma - \Sigma_0!$

original term	0 th order	1 st order	2 nd order
f_0	f_{00}	$f_{01}(\Delta\Sigma)$	$\frac{1}{2} f_{02}(\Delta\Sigma)^2$
$f_1 H$	—	$f_{10} H$	$f_{11} \Delta\Sigma H$
$f_2 H^2$	—	—	$f_{20} H^2$

Red annotations for expansion:
 $f_0 = f_{00} + f_{01} \Delta\Sigma + \frac{1}{2} f_{02} \Delta\Sigma^2 + o(\Delta\Sigma^3)$
 $f_1 = f_{10} + f_{11} \Delta\Sigma + o(\Delta\Sigma^2)$
 $f_2 = f_{20} + o(\Delta\Sigma)$

Total, we get:

$$f_{el} = f_{00} + f_{01} \Delta\Sigma + \frac{1}{2} f_{02} \Delta\Sigma^2 + f_{10} H + f_{11} \Delta\Sigma H + f_{20} H^2$$

0.

Previously:

$$f_{\text{ex}} = f_{00} + \frac{1}{2} f_{02} \Delta \Sigma^2 + f_{10} H + \underbrace{f_{11} \Delta \Sigma H}_{\text{"coupling term" between } H \text{ and } \Sigma} + f_{22} H^2$$

"coupling term"
between H and Σ .
The minimizing $\Delta \Sigma$
will depend on H .

What Safran does
Next

Let's find $\Delta \Sigma^* = \Sigma^* - \Sigma_0$, where Σ^* is the Minimizing area for finite curvature:

$$\left. \frac{\partial f_{\text{ex}}}{\partial \Delta \Sigma} \right|_{\Delta \Sigma^*} = f_{02} \Delta \Sigma^* + f_{11} H = 0$$

$$\Delta \Sigma^* = -\frac{f_{11}}{f_{02}} H$$

Q1:
← what's a sanity check
for this expression?

Assuming the membrane can adjust Σ as a function of H , the curvature free energy is:

$$f = f_{\text{ex}}(\Delta \Sigma^*) = f_{00} + \frac{1}{2} f_{02} \left(-\frac{f_{11} H}{f_{02}} \right)^2 + f_{02} H^2 - \frac{f_{11}^2}{f_{02}} H^2 + f_{10} H$$

$$f = f_{00} + f_{10} H + \left(f_{02} - \frac{f_{11}^2}{2f_{02}} \right) H^2$$

Now at this point, Safran redefines stuff

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ g_0 & g_1 & g_2 \end{array}$$

$$f = g_0 + g_1 H + g_2 H^2$$

$$g_2 \rightarrow 2k_c$$

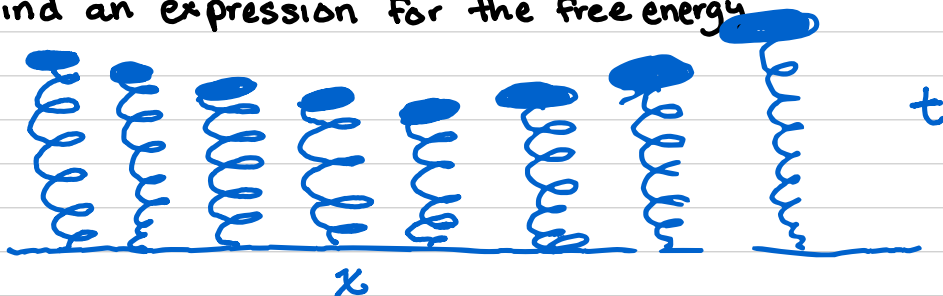
$$g_1 / 2g_2 \rightarrow c_0$$

But suppose instead of minimizing Σ , we just let it vary.

Example Application:

Lipids adhering to a surface:

Find an expression for the free energy



Each lipid has an area Σ_i , thickness t_i , curvature H_i , and we can sum over the lipids to get the total F.e

$$F_{el} = \sum_i f_{el,i} = \sum_i \left[\frac{f_{02}}{2} (\Delta \Sigma_i)^2 + f_{10} H_i + f_{11} H_i \Delta \Sigma_i + f_{20} H_i^2 \right]$$

but an integral version will be more tractable:

$$\sum_i f_{el,i} \rightarrow \int dx dy f_{el}(x,y) / \Sigma_i(x,y)$$

That is: ($\Sigma, \Delta \Sigma, H$ are all functions of x and y)

$$F_{el} = \int dx dy \left\{ f_{02} \frac{(\Delta \Sigma)^2}{\Sigma} + f_{10} \frac{H}{\Sigma} + f_{11} H \frac{\Delta \Sigma}{\Sigma} + f_{20} \frac{H^2}{\Sigma} \right\}$$

To deal with all these Σ s, use volume conservation:

$$t_0 \Sigma_0 = t \Sigma \rightarrow t_0 \Sigma_0 = [t_0 + \overbrace{(t-t_0)}^{\Delta t}] \Sigma = (t_0 + \Delta t) \Sigma$$

$$\Sigma_0 = (1 + \Delta t / t_0) \Sigma$$

Identities and Approximations

<u>Term</u>	<u>Equivalent term (to 2nd order)</u>
Σ_0	$(1 + \Delta t/t_0)\Sigma$ volume conservation
$1/\Sigma$	$(1 + \Delta t/t_0)/\Sigma_0$ rearrangement
$\Delta\Sigma$	$(\Delta t/t_c)\Sigma$ $\Delta\Sigma = \Sigma - \Sigma_0$
$\frac{(\Delta\Sigma)^2}{\Sigma}$	$(\Delta t/t_0)^2 \Sigma_0$ drop 3 rd order terms like $(\Delta t^3 \Delta\Sigma)$
H/Σ	$\frac{1}{2} (\nabla^2 t) (1 + \Delta t/t_0)/\Sigma_0$ Monge gauge
$\frac{H\Delta\Sigma}{\Sigma}$	$\frac{1}{2} (\nabla^2 t) (\Delta t/t_0)$
$\frac{H^2}{\Sigma}$	$\frac{1}{4} \frac{(\nabla^2 t)^2}{\Sigma_0}$ Drop 3 rd order terms like $(\nabla^2 t)^2 \Delta t$

Exercise: Verify each identity

Using the substitutions/identities in the previous table,

$$F = \int dx dy \left\{ f_{0,2} \frac{(\Delta \Sigma)^2}{\Sigma} + f_{1,0} \frac{H}{\Sigma} + f_{1,1} \frac{H \Delta \Sigma}{\Sigma} + f_{2,0} \frac{H^2}{\Sigma} \right\}$$

becomes

$$F = \int dx dy \left\{ f_{0,2} \frac{\Sigma_0 (\Delta t)^2}{t_0^2} + \frac{f_{1,0}}{2 \Sigma_0} \nabla^2 t \left(1 + \frac{\Delta t}{t_0}\right) + f_{1,1} \frac{\nabla^2 t}{2} \frac{\Delta t}{t_0} + \frac{f_{2,0}}{4} \frac{(\nabla^2 t)^2}{\Sigma_0} \right\}$$

or, rearranging,

$$F = \int dx dy \left\{ f_{0,2} \frac{\Sigma_0 (\Delta t)^2}{t_0^2} + \frac{f_{1,0}}{2 \Sigma_0} \nabla^2 t + \frac{1}{2 \Sigma_0 t_0} (f_{1,0} \otimes \Sigma_0 f_{1,1}) \nabla^2 t \Delta t + \frac{f_{2,0}}{4} \frac{(\nabla^2 t)^2}{\Sigma_0} \right\}$$

should be minus
not sure yet where I
lost it

So we can give all these constants more traditional names:

$$f_{0,2} \Sigma_0 \rightarrow \frac{k_A}{2} \quad \frac{f_{1,0}}{\Sigma_0} \rightarrow 2k_c c_0 \quad \frac{f_{1,1}}{\Sigma_0} \rightarrow 2k_c c'_0 \quad \frac{f_{2,0}}{\Sigma_0} \rightarrow k_c$$

so,

area derivative
of c_0

$$F = \int dx dy \left\{ \frac{k_A}{2} \left(\frac{\Delta t}{t_0} \right)^2 + k_c c_0 \nabla^2 t + k_c (c_0 - c'_0 \Sigma_0) \nabla^2 t \left(\frac{\Delta t}{t_0} \right) + \frac{k_c}{4} (\nabla^2 t)^2 \right\}$$

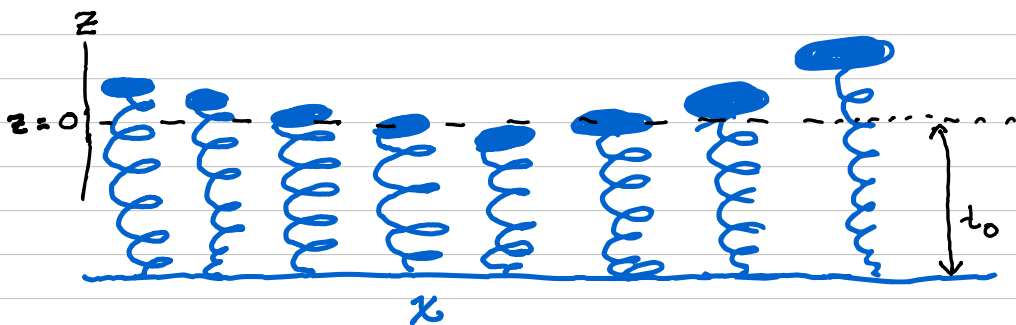
define $\ell_2 = c_0 - c'_0 \Sigma_0$
and that's where ℓ comes from!

So now we have an expression just in terms of t !

Well, actually, it's in terms of $\nabla^2 t$ and Δt .

But $\nabla^2 t = \nabla^2(\Delta t)$! So we can put the whole expression in terms of Δt . But $\nabla^2(\Delta t)$ looks terrible. Let's give Δt a new name, z , and then:

$$F = \int dx dy \left(\frac{k_A}{2} \frac{z^2}{t_0^2} + k_c c_0 \nabla^2 z + k_c \int \frac{z}{t_0} \nabla^2 z + \frac{k_c}{4} (\nabla^2 z)^2 \right)$$



Final thought... why $\xi = c_0 - c'_0 \Sigma_0$ instead of just Σ_0 ? What will happen if ξ is negative?